

Data lakehouse implementation best practices

So far, you've learned that data lakehouse architecture combines a data lake with a data warehouse to take advantage of the benefits of both data storage solutions. You've also explored the five layers of data lakehouse architecture, which are the building blocks for any lakehouse architecture. These layers include: ingestion, storage, metadata, API, and consumption. In this reading, you'll explore how cloud data professionals approach lakehouse implementation, and learn some best practices for the future.

Best practices for data lakehouse architecture

Data lakehouses are a newer solution; the lakehouse innovates on the traditional data warehouse and the data lake to create a storage solution suited to cloud-based systems. As a cloud data professional, this means you might have the opportunity to implement a data lakehouse architecture for your organization. When designing a lakehouse architecture, there are a few key considerations that will guide your implementation:

- **Security:** As with any data storage solution, security should be a high priority. You want to make sure that data is safe and secure when it is being stored, and when it is being processed.
- **Performance:** Performance is one of the great benefits of a lakehouse architecture. Optimizing performance as part of the design process by partitioning data and using indexes will make your system as high-performing as possible.
- **Scalability and flexibility:** Because you're working with cloud systems, scalability and flexibility are a must; your architecture should be able to handle current data needs and be adaptable to future needs.
- **Governance:** Data governance ensures the accuracy, reliability, and security of an organization's data. Setting up effective governance will help maintain the lakehouse's usability long-term.
- **Self-service:** One of the benefits of a cloud-based system is that authorized users can access the data they need anytime and anywhere. Prioritizing the user's ability to use the system to answer their questions also makes the system more beneficial to users overall.

Key takeaways

Lakehouse architecture is a great solution that enables cloud data analytics professionals to take advantage of the benefits of both data warehouses and data lakes, while compensating for their drawbacks. Understanding how professionals approach designing data lakehouse architecture will help you better understand how these systems function, and help you implement your own architecture in the future.